Likert (lick-urt) or lay it:

using ordinal regression for modeling clutch size &| similar things







Examples of ordinal data

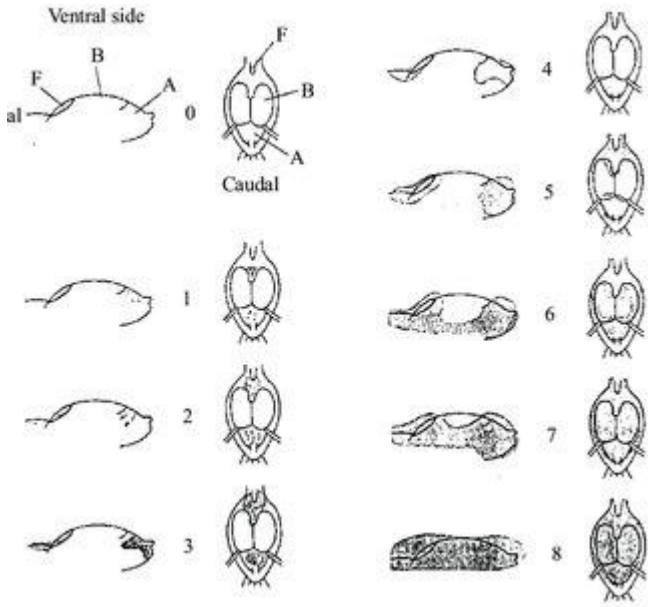
Credit Rating Scales by Agency, Long-Term

Moody's	S&P	Fitch	
Aaa	AAA	AAA	Prime
Aa1	AA+	AA+	High grade
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non-investment grade speculative
Ba2	BB	BB	
Ba3	BB-	BB-	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC		Extremely speculative
Caa3	CCC-		Default imminent with little prospect for recovery
Ca	CC	CC	
	C	C	
C	D	D	In default
/			
/			

"Junk"

WOLFSTREET.com







These are simulated (and real) data!

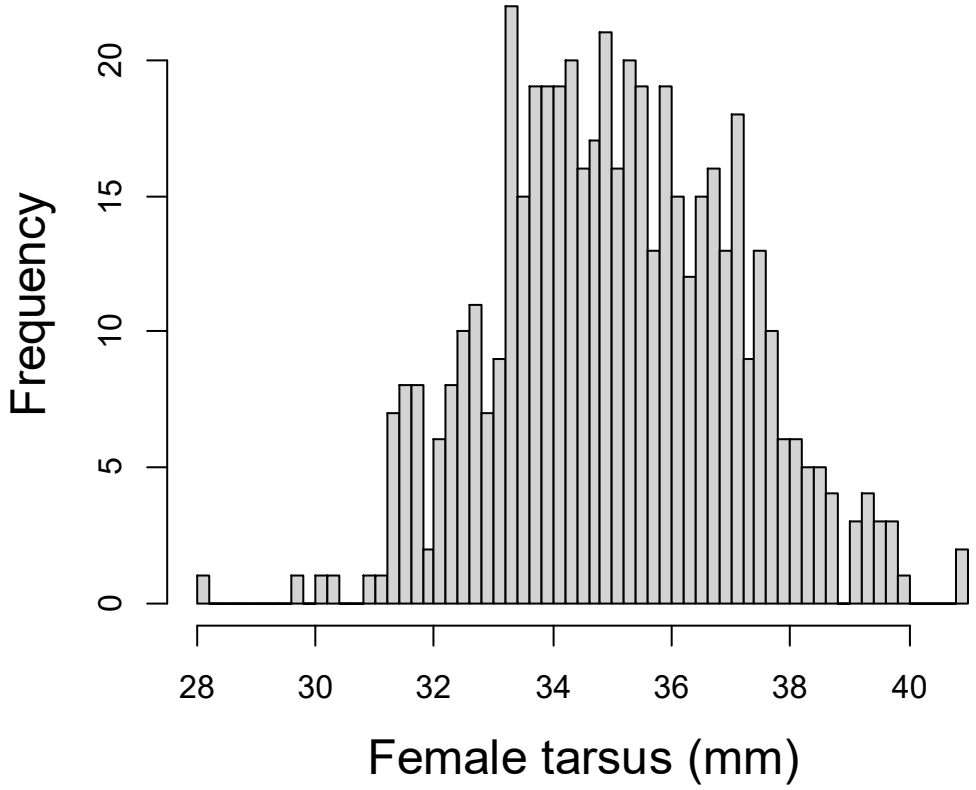


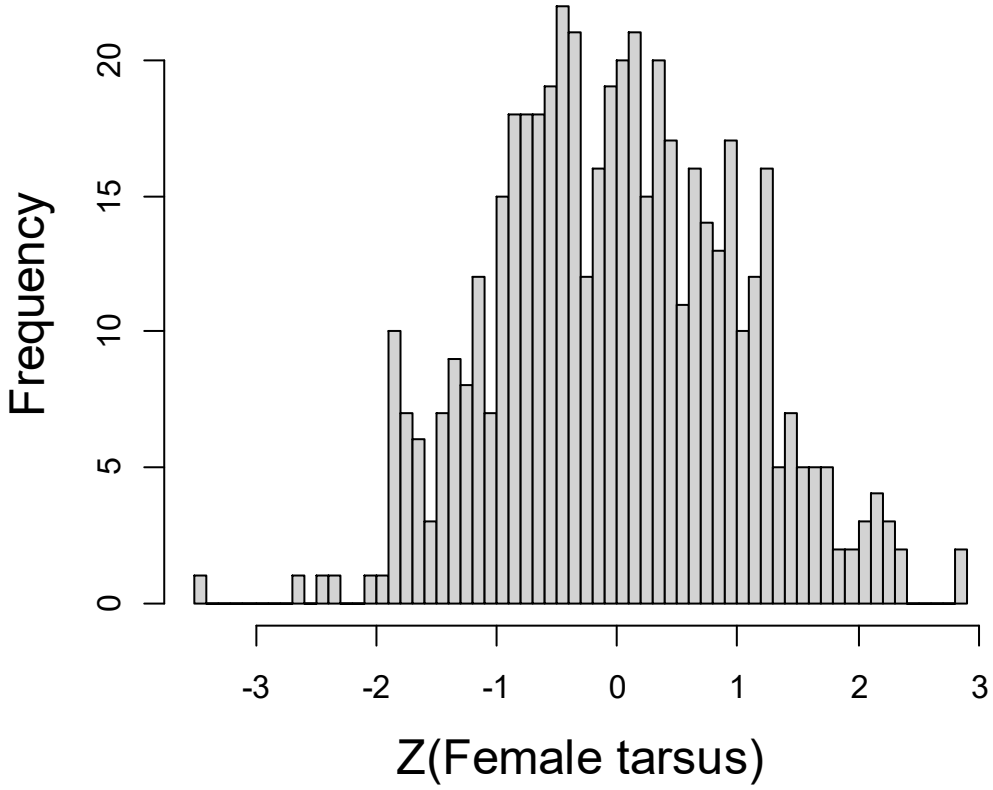
Simulated generic bird (5-9 eggs)
(*Birdus birdus*)



Pacific black brant (2-6 eggs)
(*Branta bernicla nigricans*)

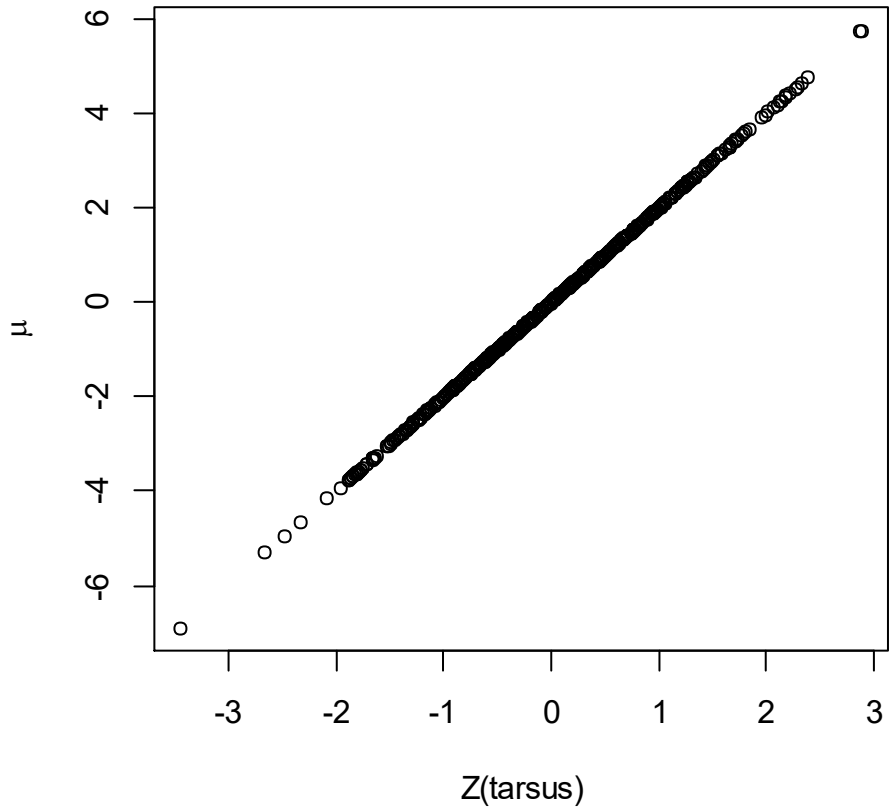
Simulated data: simulate variation in size



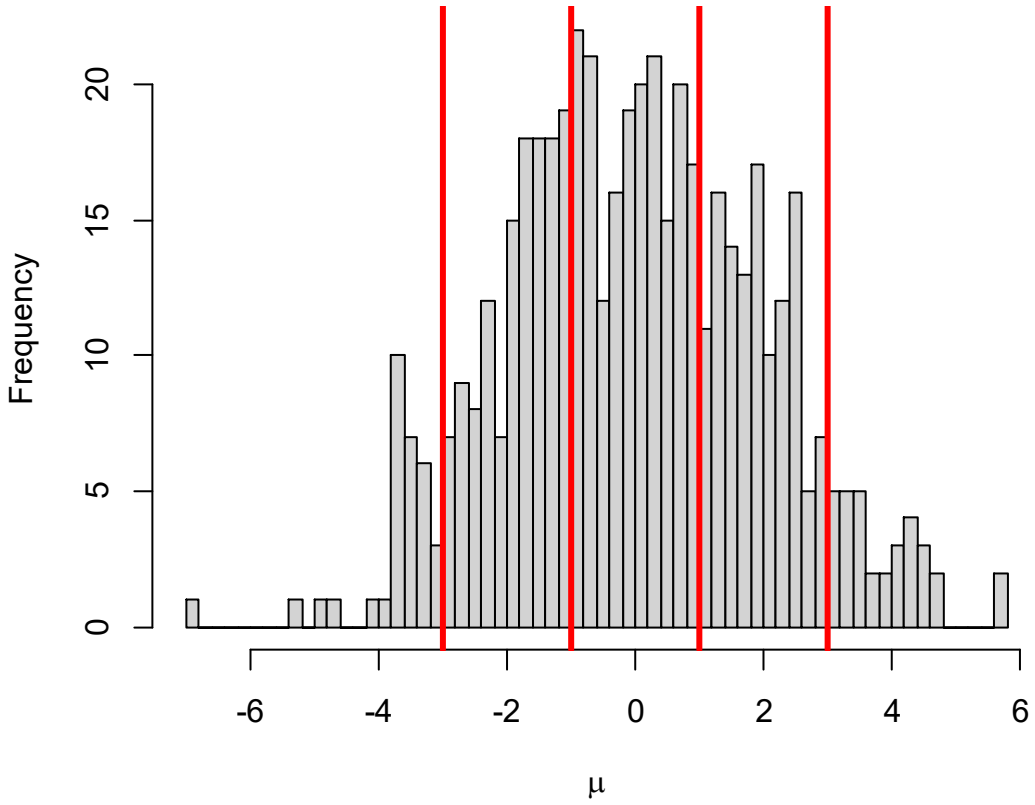


Simulated data: simulate regression coefficient, cutpoints

$$\mu_i = \zeta \times z_i$$
$$\zeta = 2$$
$$c = [-3, -1, 1, 3]$$



Simulated data: derive clutch size probabilities

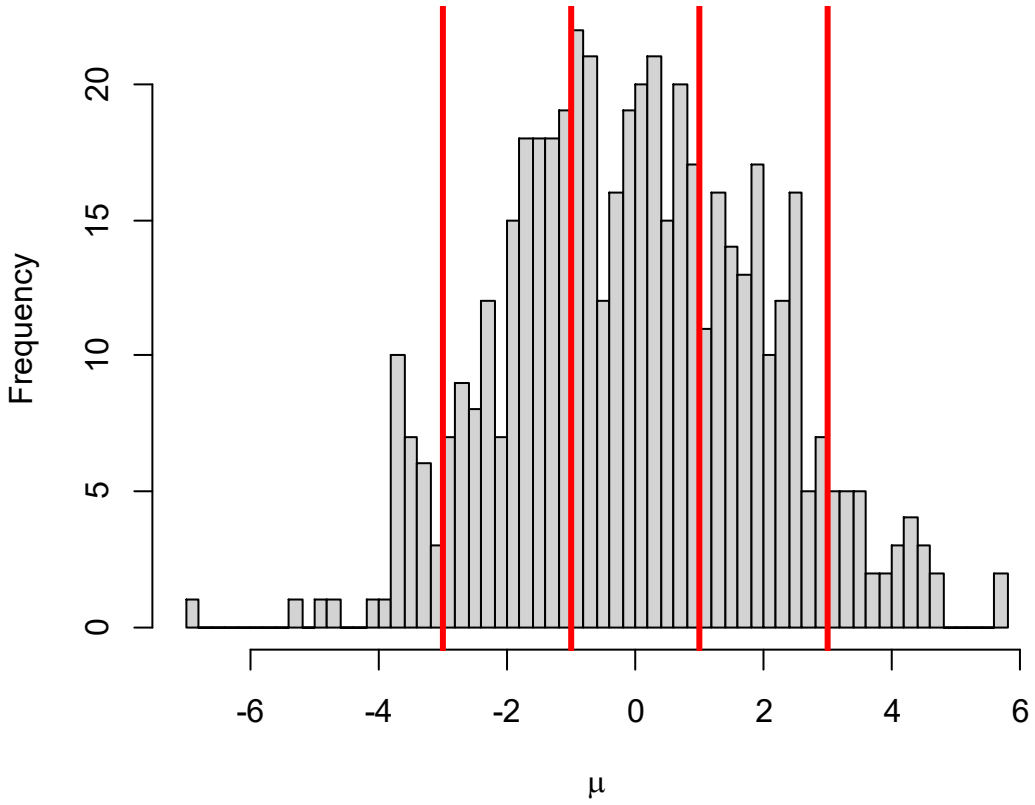


$$c = [-3, -1, 1, 3]$$

$$q_{i1} = \text{logit}^{-1}(c_1 - \mu_i)$$

$$q_{i1} = \text{logit}^{-1}(c_1 - \mu_i) = p_{i1}$$

Simulated data: derive clutch size probabilities

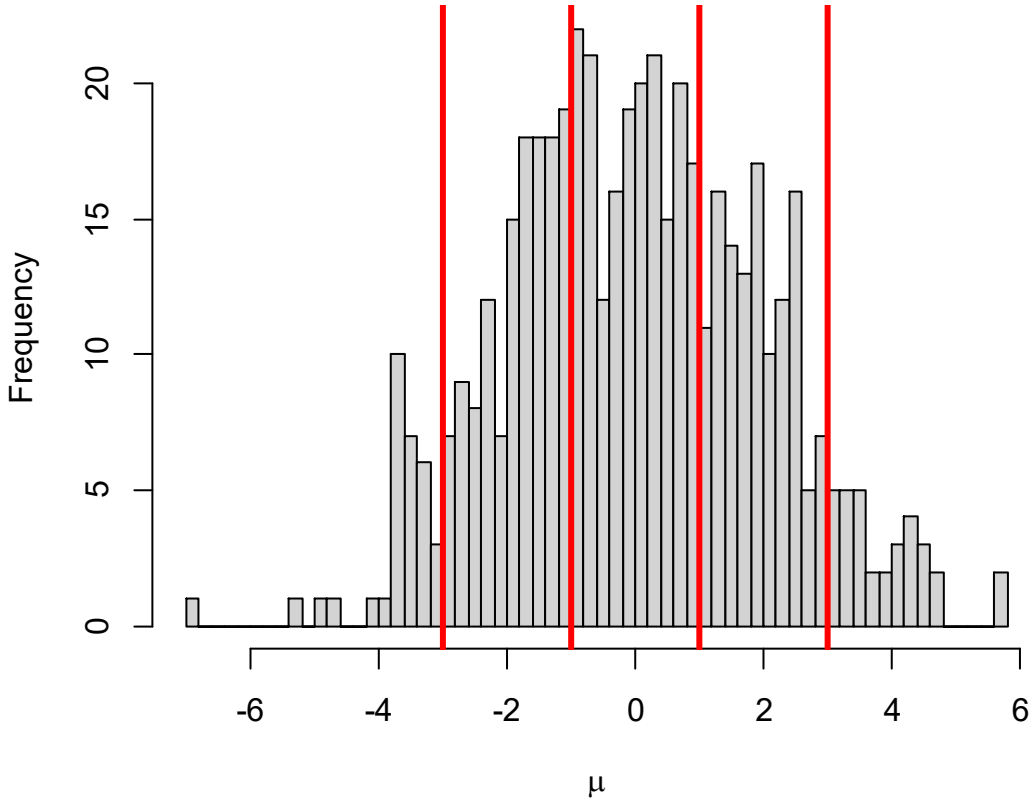


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Simulated data: derive clutch size probabilities

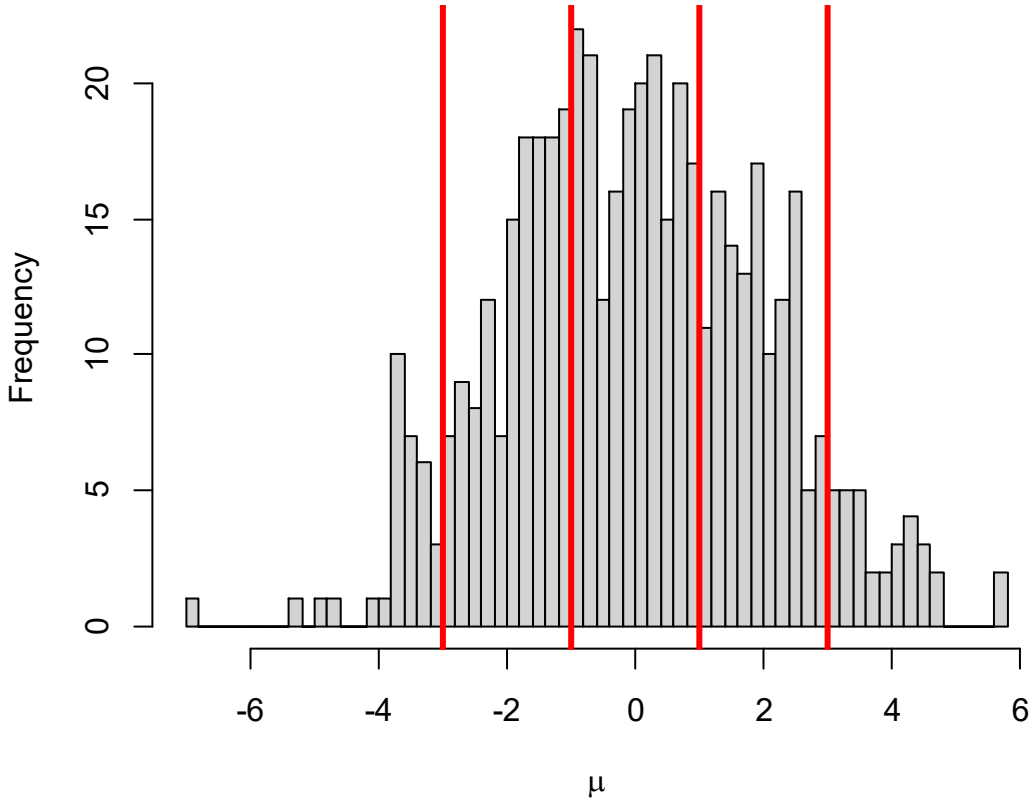


$$c = [-3, -1, 1, 3]$$

$$q_{ij} = \text{logit}^{-1}(c_j - \mu_i)$$

$$p_{ij} = q_{ij} - q_{ij-1}$$

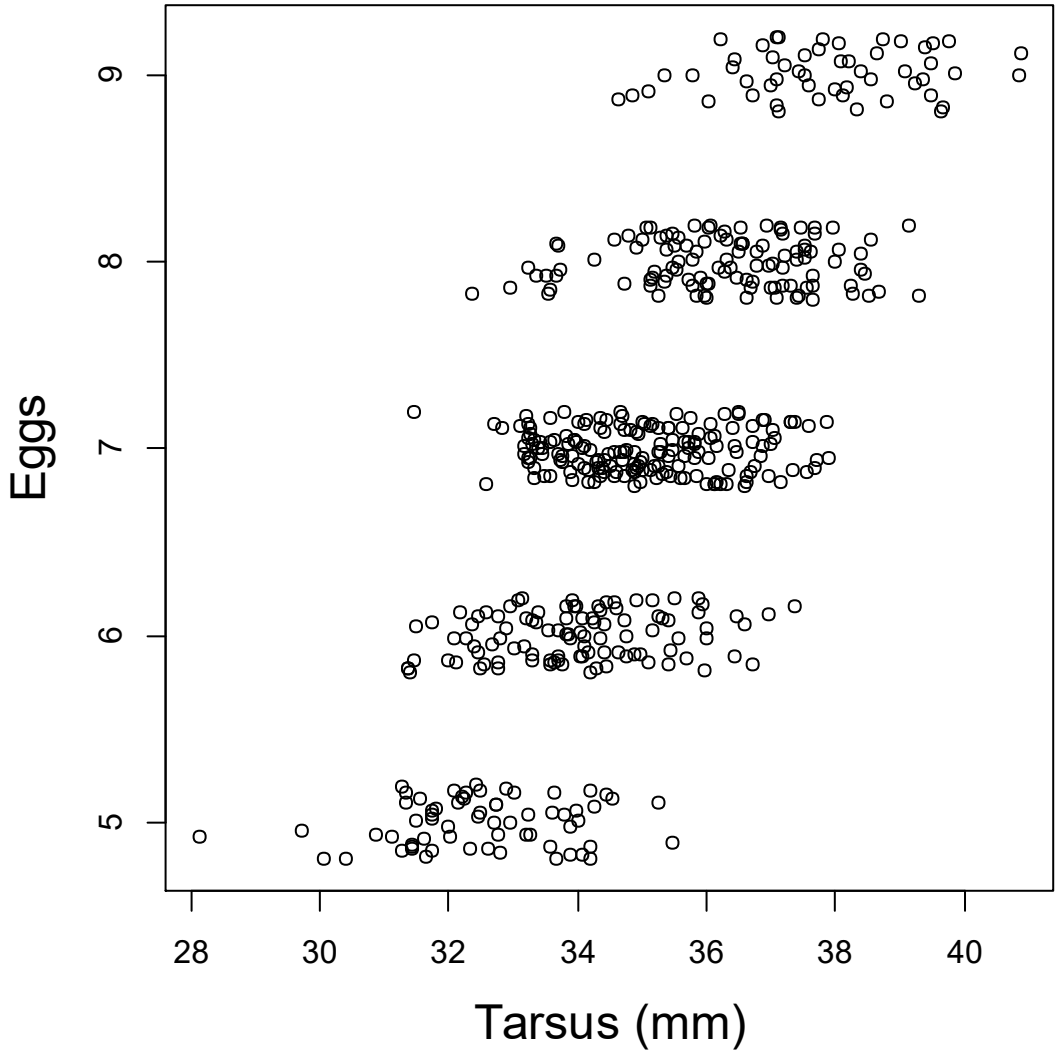
Simulated data: derive clutch size probabilities



$$c = [-3, -1, 1, 3]$$

$$p_{iJ} = 1 - q_{iJ-1}$$

Simulated data: simulate clutch sizes



$$y_i \sim \text{categorical}(\boldsymbol{p}_i)$$

Simulated data: the model (Poisson)

$$\ln(\mu_i) = \alpha + \beta \times x_i$$

$$y_i \sim \text{Poisson}(\mu_i)$$

$$y_i^* \sim \text{Poisson}(\mu_i)$$

Simulated data: the model (ordinal logistic)

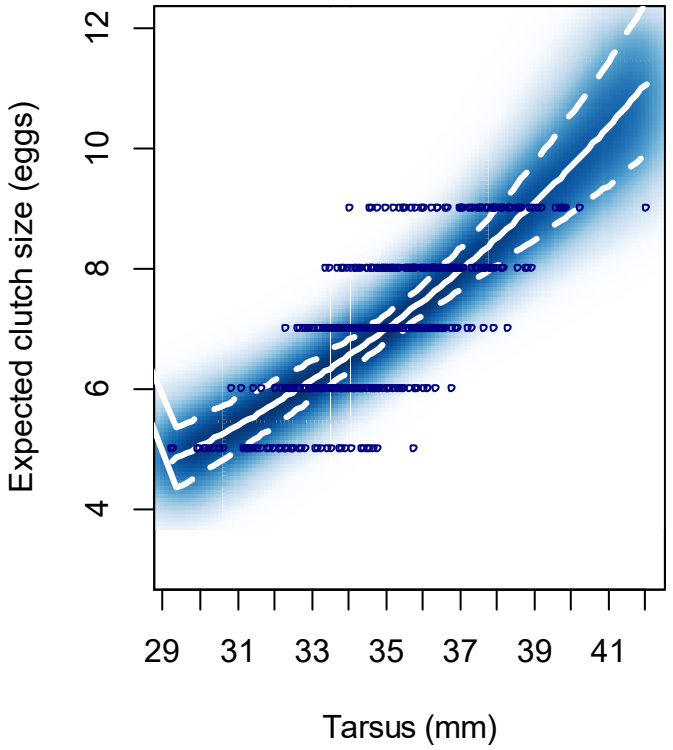
$$\mu_i = \beta \times x_i$$

$$y_i \sim \text{ordered.logistic}(\mu_i, \boldsymbol{\alpha})$$

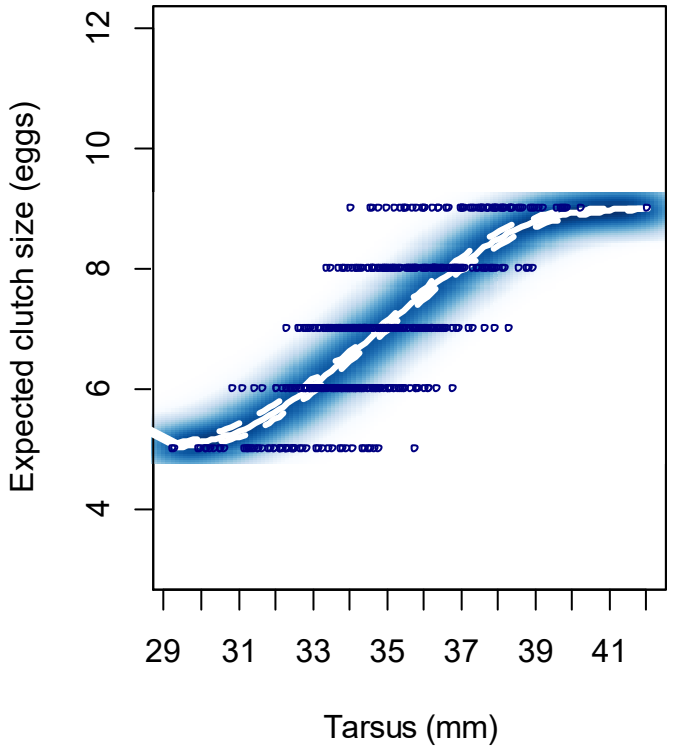
$$y_i^* \sim \text{ordered.logistic}(\mu_i, \boldsymbol{\alpha})$$

Expected values and the real data

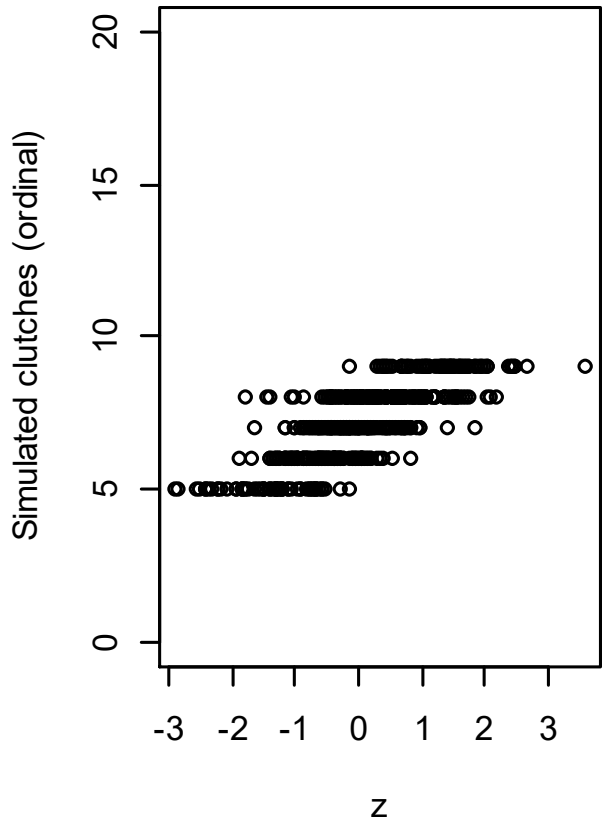
Poisson regression

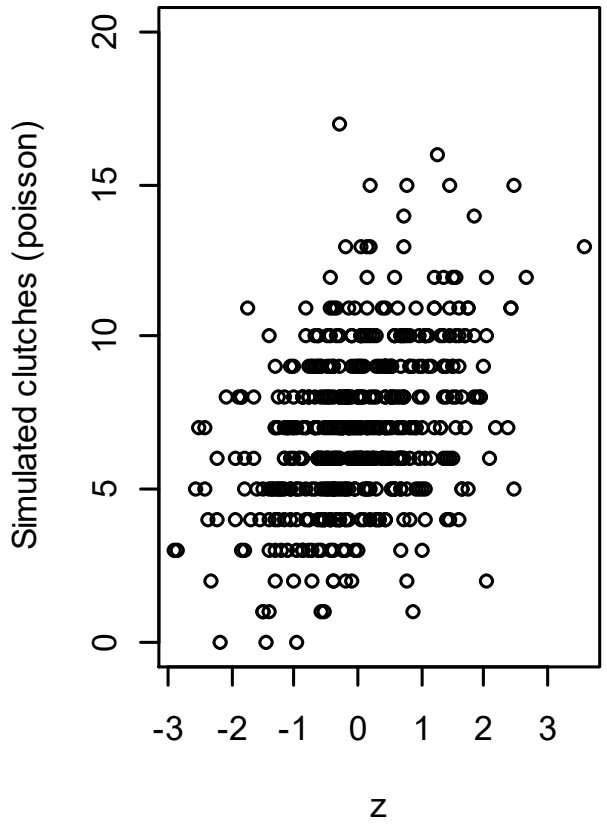


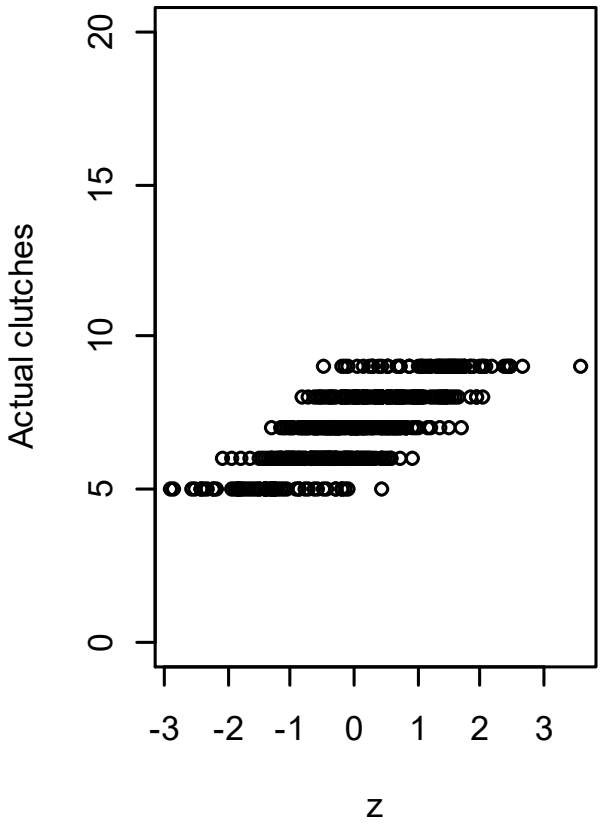
Ordinal regression



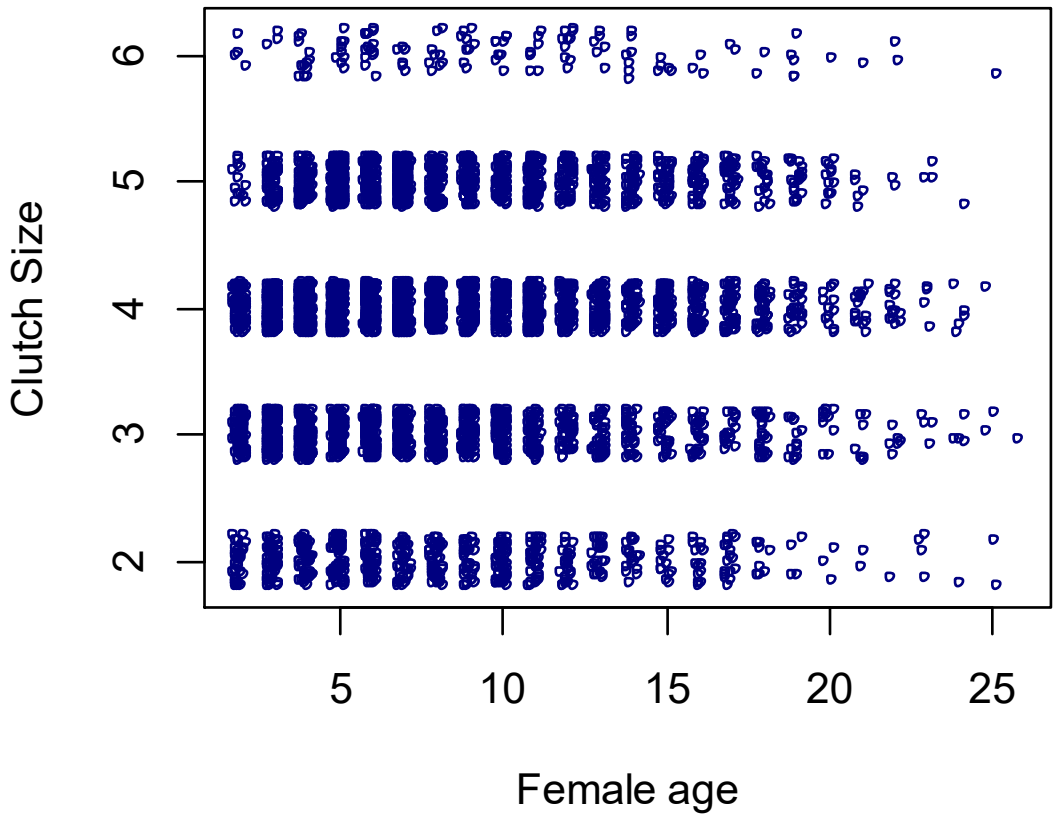
Goodness-of-fit...





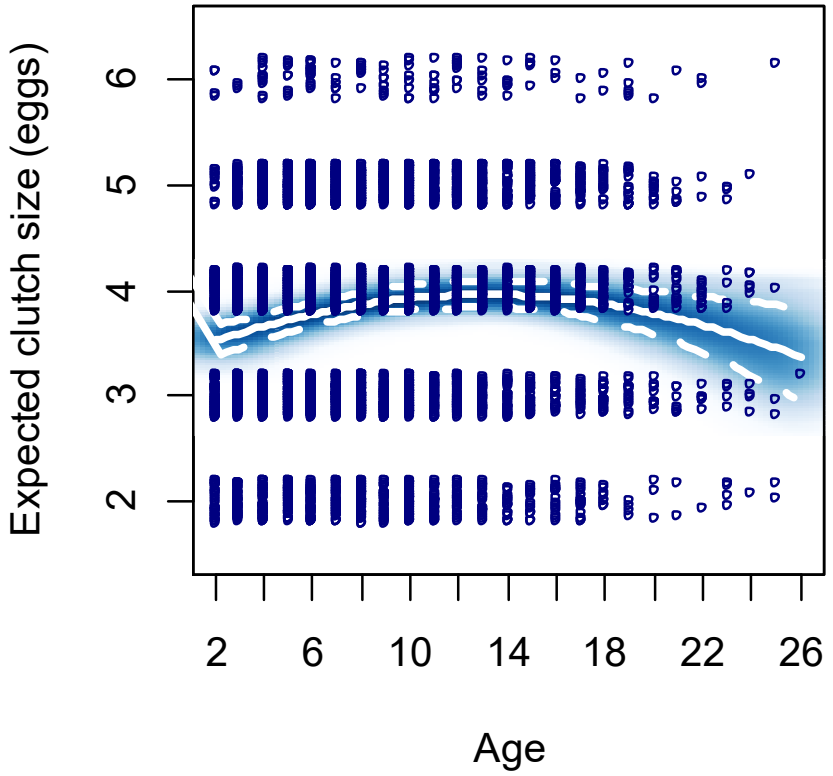


Real data: the data

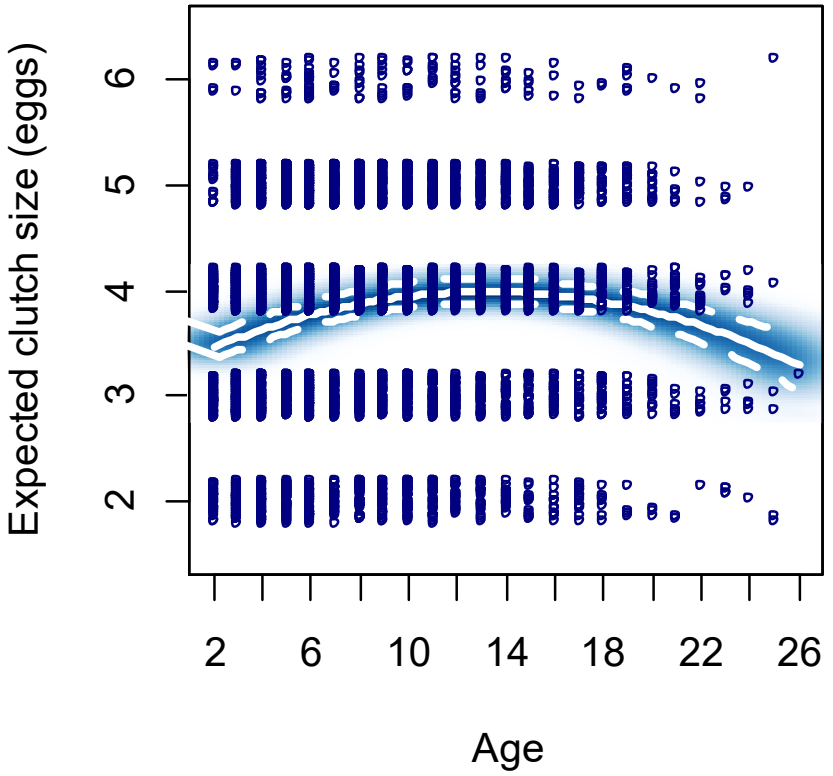


Expected values and the real data

Poisson regression



Ordinal regression



Exploring fit: the simulated data

